

MONTANA, Switzerland

WMO: 067240

Lat: 46.2986N

Lon: 7.4608E

Elev: 4948

StdP: 12.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	13.8	-3.2	5.6	17.0	1.5	7.1	21.1	17.3	36.2	15.2	35.0	2.8	30	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	77.3	59.1	74.2	58.1	71.4	57.2	60.9	73.4	59.5	71.1	58.2	68.9	6.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.4	81.5	64.0	54.9	77.3	63.0	53.7	73.7	62.4	29.4	73.7	28.4	71.5	27.4	69.0	66.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.0	11.2	DB	6.0	81.8	4.9	3.1	2.4	84.0	-0.5	85.8	-3.2	87.5	-6.8	89.8	(4)
(5)			WB	4.5	63.6	4.7	1.3	1.1	64.5	-1.6	65.3	-4.2	66.0	-7.6	66.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	28.8	29.5	35.6	42.0	49.3	56.4	59.6	59.0	52.3	46.3	36.3	30.9	
	DBStd	13.12	7.10	8.07	7.38	7.12	6.56	7.04	6.19	6.39	6.27	6.83	8.06	7.52	
	HDD50	3255	656	573	446	253	92	21	6	7	47	150	412	592	
	HDD65	7737	1121	993	911	690	486	268	185	204	381	580	861	1057	
	CDD50	1040	0	0	0	12	71	214	304	286	117	35	1	0	
	CDD65	47	0	0	0	0	0	10	19	18	0	0	0	0	
	CDH74	279	0	0	0	0	2	65	107	104	1	0	0	0	
	CDH80	24	0	0	0	0	0	7	7	10	0	0	0	0	
Wind	WSAvg	4.1	3.8	4.0	4.6	5.1	4.6	4.1	4.0	3.6	3.5	3.9	4.2	3.9	
Precipitation	PrecAvg	41.2	3.0	3.0	2.6	2.8	3.9	4.2	4.6	4.4	3.0	2.7	3.3	3.7	
	PrecMax	54.9	9.4	12.2	7.7	5.7	7.3	8.0	9.2	7.4	5.6	6.3	8.5	10.2	
	PrecMin	32.6	0.4	0.4	0.7	0.9	0.9	2.0	2.3	1.2	1.4	0.1	0.0	0.0	
	PrecStd	5.9	2.2	2.7	1.8	1.2	1.8	1.4	1.6	1.6	1.3	1.4	2.1	2.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.7	51.0	56.7	64.8	72.4	80.0	80.6	81.3	72.3	65.9	56.8	49.5	
		MCWB	36.9	37.8	42.4	47.7	54.5	59.1	60.4	59.7	57.1	51.5	43.7	36.9	
	2%	DB	43.4	46.5	53.0	60.1	67.3	75.5	77.1	76.9	69.0	61.8	52.4	45.6	
		MCWB	34.5	36.1	40.6	46.0	52.4	58.2	59.1	59.0	55.8	49.2	42.0	36.1	
	5%	DB	40.5	42.7	49.6	56.5	63.5	71.9	73.8	73.5	65.3	58.6	49.5	43.0	
		MCWB	33.8	34.4	39.1	43.9	50.4	57.0	58.3	58.1	54.1	48.1	40.9	35.3	
	10%	DB	38.3	39.6	46.1	52.9	60.1	68.5	70.8	70.1	62.1	55.8	46.9	40.7	
		MCWB	32.7	32.7	37.2	42.2	48.6	55.5	57.3	57.2	52.8	47.0	39.6	33.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	40.0	44.1	48.8	56.0	61.7	62.7	62.5	58.9	53.6	46.4	40.6
			MCDB	43.3	47.1	54.3	62.1	69.0	75.1	75.2	76.0	68.5	61.5	52.5	44.6
2%		WB	37.1	37.3	41.6	46.6	53.4	59.8	61.0	60.5	56.8	51.4	43.9	38.3	
		MCDB	41.0	43.4	50.4	58.4	64.6	72.7	73.5	72.5	66.2	58.6	50.2	42.6	
5%		WB	35.0	35.4	39.8	44.7	51.4	57.9	59.5	59.2	55.1	49.7	42.0	36.7	
		MCDB	39.1	41.4	47.6	54.4	61.2	69.4	71.1	70.4	63.4	56.2	47.7	41.1	
10%		WB	33.2	33.4	38.1	42.9	49.7	56.3	58.1	57.9	53.5	48.1	40.3	35.1	
		MCDB	37.4	38.9	44.7	51.5	58.5	66.6	68.6	68.1	60.7	54.3	45.8	39.4	
Mean Daily Temperature Range		5% DB	MDBR	10.1	11.9	13.8	14.8	15.6	17.0	17.4	16.4	15.0	13.2	10.4	10.1
			MCDBR	12.2	15.1	17.5	19.1	19.9	20.8	20.9	20.6	18.2	16.2	13.2	12.8
	MCWBR		8.0	9.4	9.9	9.8	9.7	9.8	9.4	9.1	9.6	9.3	8.0	7.9	
	5% WB	MCDBR	10.9	13.8	15.7	18.1	18.3	19.5	19.4	18.8	16.5	14.6	12.2	10.8	
		MCWBR	8.1	9.4	9.3	9.6	9.6	9.7	9.6	9.3	9.3	9.1	7.8	7.7	
Clear-Sky Solar Irradiance	taub	0.237	0.261	0.304	0.343	0.361	0.365	0.363	0.351	0.328	0.305	0.268	0.237		
	taud	2.341	2.285	2.237	2.245	2.283	2.316	2.316	2.361	2.417	2.463	2.453	2.379		
	Ebn at Noon	280	292	292	289	287	285	284	284	282	273	266	268		
	Edh at Noon	23	30	37	41	41	40	39	36	31	25	21	20		
All-Sky Solar Radiation	RadAvg	455	732	1085	1390	1585	1885	1873	1632	1279	830	473	373		
	RadStd	52	101	150	191	135	171	155	138	108	88	65	56		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.62	+0.74	+0.49	N/A	N/A	+108	+18
(47) Regional (109 neighbors)		N/A	N/A	N/A	+1.28	+0.72	+0.54	N/A	N/A	+103	+29

Nomenclature: See separate page